

Trade Policy and Welfare

A study companion in six parts, with lesson quizzes and a comprehensive final exam

Lecture Notes

Roadmap

These notes cover the welfare economics of trade policy in six parts: (I) partial-equilibrium effects of tariffs and quotas in small and large countries; (II) the theory of the second best when domestic market power is present, plus the game-theory toolkit (best responses, strategic substitutes/complements, Cramer's rule, the envelope theorem); (III) a Cournot duopoly model of strategic trade policy; (IV) the classic theory of *optimum tariffs and retaliation* — Johnson (1953–54), Mayer (1981), Syropoulos (2002); (V) strategic trade and industrial policy under imperfect competition, following Brander and Spencer (1984, 1985), with the Cournot-versus-Bertrand caveat on when the optimal export policy flips from a subsidy to a tax; and (VI) the political economy of trade policy — Mayer (1984) and Grossman–Helpman (1994) — explaining why actual protection deviates from the welfare-maximizing benchmarks. Every lesson ends with an understanding quiz; a comprehensive final exam and full answer key close the notes.

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Part I

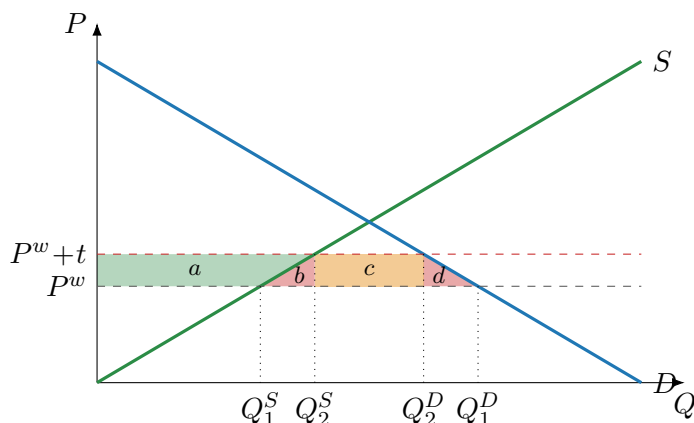
Partial-Equilibrium Trade Policy

The instruments: **tariffs**, **customs** (non-tariff administrative barriers), **export subsidies**, and **quotas**. The questions: what are their welfare effects (this part); what are the *optimal* policies (Parts II–V); what happens under *strategic interaction* — tariff wars and retaliation (Part IV); and how do **trade agreements** resolve the strategic tariff-setting game cooperatively (Part IV). This is the organizing logic of the whole course: from a single country taking the rest of the world as given, to a single country with *market power* interacting with foreign competitors who react optimally, to the fully strategic problem of *two* policy-setting governments — and finally to governments that are not benevolent at all.

1 Lesson 1: The small-country tariff

1.1 Setup and welfare decomposition

A country is “small” if it faces a perfectly elastic (horizontal) world supply of the imported good at price P^w . Let $D(P)$ and $S(P)$ be domestic demand and supply, and let t be a specific tariff, so the domestic price becomes $P^w + t$. In the figure, drawn with $S : Q = 2P$ and $D : Q = 10 - 2P$, $P^w = 1.5$ and $t = 0.5$: imports shrink from $Q_1^D - Q_1^S = 7 - 3 = 4$ to $Q_2^D - Q_2^S = 6 - 4 = 2$.



Welfare decomposition (small country)

$$\Delta CS = -(a + b + c + d), \quad \Delta PS = +a, \quad \Delta TR = +c,$$

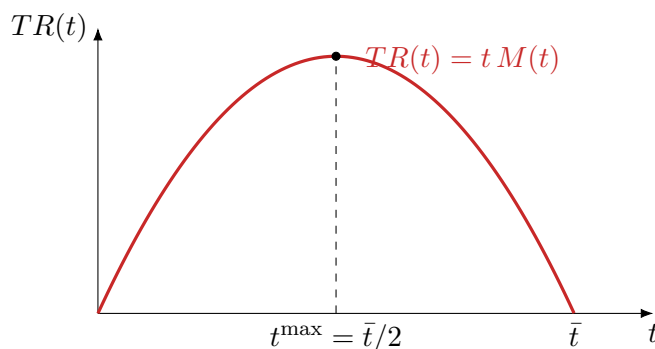
$$\Delta W = \Delta CS + \Delta PS + \Delta TR = -(b + d) < 0.$$

A tariff in a small country is unambiguously welfare-reducing: it only creates deadweight loss (the production distortion b and the consumption distortion d) because the country cannot improve its terms of trade — the world price P^w is fixed.

1.2 Tariff revenue: the Laffer curve

Tariff revenue $TR(t) = t \times M(t)$ is zero at $t = 0$ (a wedge on a large trade volume, but zero rate) and zero again at the **prohibitive tariff** $\bar{t}$ (a high rate on zero imports). In between it is

hump-shaped. With linear import demand $M(t) = M_0(1 - t/\bar{t})$, revenue $TR = t M_0(1 - t/\bar{t})$ peaks exactly at $t^{\max} = \bar{t}/2$.



Note that the *revenue*-maximizing tariff is not the *welfare*-maximizing tariff: for a small country the latter is zero, and even for a large country (Lesson 2) the optimum tariff lies strictly below $t^{\max}$, because welfare counts the deadweight losses that revenue ignores.

Understanding Quiz 1 — Lesson 1

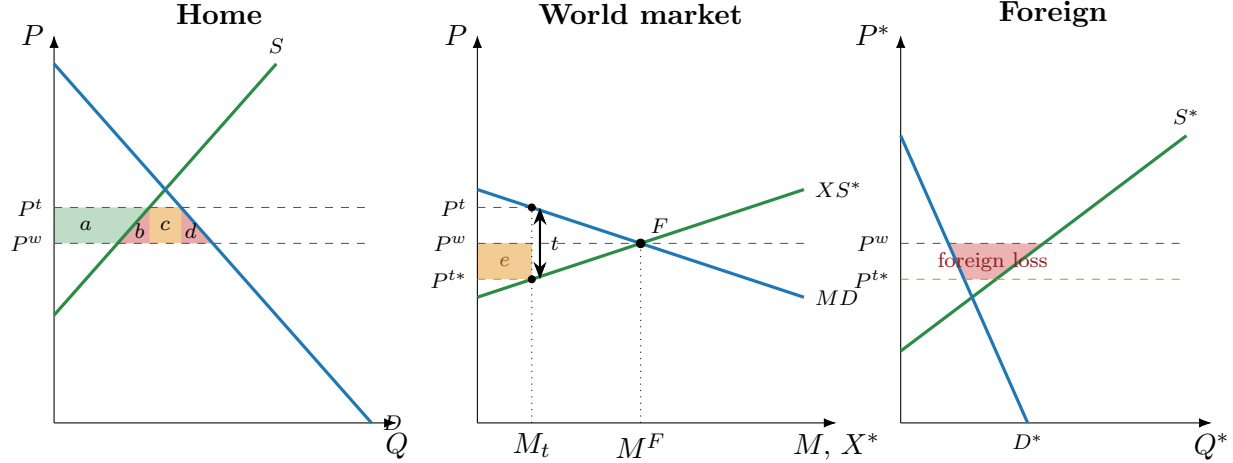
- Q1.1** In the small-country tariff diagram, the areas b and d measure, respectively:
- (A) producer gain and government revenue
 - (B) the production distortion (too much high-cost domestic output) and the consumption distortion (too little consumption)
 - (C) terms-of-trade gain and quota rent
 - (D) transfers to foreigners
- Q1.2** The net welfare effect of a small-country tariff is:
- (A) $+c$
 - (B) $-(b + d) < 0$: unambiguously negative
 - (C) $+a - c$
 - (D) zero
- Q1.3** Tariff revenue is zero at both $t = 0$ and $t = \bar{t}$ because:
- (A) the government forgets to collect it
 - (B) at $t = 0$ the rate is zero; at $\bar{t}$ imports are zero — revenue is rate $\times$ volume
 - (C) demand is perfectly inelastic
 - (D) the world price adjusts
- Q1.4** Why can't a small country gain from any tariff?
- (A) its government is too weak
 - (B) it faces a fixed world price, so there is no terms-of-trade gain to offset the deadweight losses
 - (C) its consumers do not value imports
 - (D) tariffs are illegal

2 Lesson 2: The large-country tariff and the optimum tariff

2.1 Three-panel analysis

If the country is “large,” its tariff drives a wedge between the price paid domestically, P^t , and the (lower) price received by foreign exporters, P^{t*} , with $P^t - P^{t*} = t$. The three-panel diagram

tracks Home, the world market (Home's import demand $MD = D - S$ meeting Foreign's export supply $XS^* = S^* - D^*$), and Foreign. The figure is drawn to be internally consistent: Home has $D : Q = 10 - 2P$, $S : Q = 2P - 3$, so $MD : M = 13 - 4P$; Foreign has $S^* : Q = 3P - 3$, $D^* : Q = 4 - P$, so $XS^* : M = 4P - 7$. Free trade: $P^w = 2.5$, $M = 3$. A tariff $t = 1$ gives $P^t = 3$, $P^{t*} = 2$, $M = 1$.



The tariff moves the world-market equilibrium from F down along XS^* : imports contract from M^F to M_t , the buyer's price rises to P^t , and the seller's price falls to P^{t*} . The shaded rectangle e in the world panel is revenue collected at Foreign's expense; the shaded band in the Foreign panel is the net loss of Foreign surplus (of which e is the transfer to Home and the remainder Foreign's own deadweight loss).

Large-country tariff decomposition

Relative to free trade at P^w : Home consumers face a higher price P^t (losing surplus), Home producers gain, the government collects revenue $t \times M = (c + e)$ on the new import volume, and — crucially — the price *received* by foreign exporters falls to $P^{t*} < P^w$. The part of revenue collected “from foreigners,” $e = (P^w - P^{t*}) M$, is a **terms-of-trade gain**: a pure transfer from Foreign. Hence

$$\Delta CS = -(a + b + c + d), \quad \Delta PS = +a, \quad \Delta TR = +(c + e), \quad \boxed{\Delta W = -b - d + e,}$$

which is *positive* for a sufficiently small t (e is first-order in t , while $b + d$ is second-order) — the classical **optimum tariff argument** (Bickerdike–Kaldor–Johnson). For a small country $e = 0$ and we are back to Lesson 1.

2.2 The optimal tariff formula

Let ε_2^* denote the price elasticity of the *foreign export supply* curve (equivalently, of Foreign's offer curve) at the equilibrium. Maximizing Home welfare ($E_u du = -M_2 d\pi + \pi dM_2$, the envelope condition on Home's expenditure function) with respect to the tariff yields:

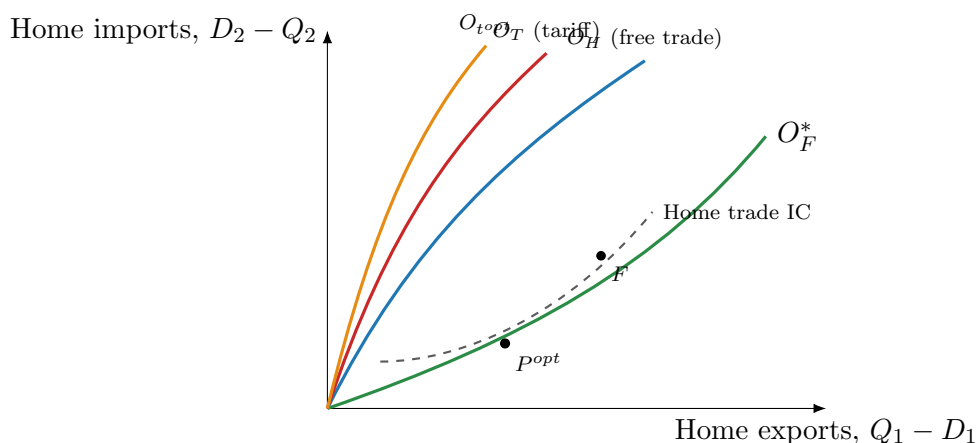
Johnson's optimum tariff formula

$$t^{opt} = \frac{1}{\varepsilon_2^*}, \quad \varepsilon_2^* = \xi_1^* - 1,$$

with ξ_1^* the elasticity of Foreign's offer curve. The optimal tariff is *inversely* related to the elasticity of the trade partner's export supply: the less elastic Foreign's willingness to supply exports, the higher is Home's optimal (terms-of-trade-improving) tariff. If Foreign's offer curve is infinitely elastic (the small-country benchmark), $\varepsilon_2^* \rightarrow \infty$ and $t^{opt} \rightarrow 0$: **a small country's optimal unilateral tariff is zero.**

2.3 Offer-curve representation

In offer-curve (reciprocal-demand) space, Home's exports of good 1 are on the horizontal axis and its imports of good 2 on the vertical. A tariff rotates Home's offer curve toward its import axis; the **optimal** tariff selects the point on *Foreign's* offer curve O_F^* that lies on Home's highest attainable trade indifference curve — a tangency between a Home indifference curve and O_F^* .



Each successively steeper Home offer curve ($O_H \rightarrow O_T \rightarrow O_{t^{opt}}$) corresponds to a higher tariff and moves the equilibrium down along O_F^* , improving Home's terms of trade (more imports per unit of exports) but shrinking trade volume; the optimum balances the two at the tangency of a Home trade indifference curve with O_F^* .

Understanding Quiz 2 — Lesson 2

Q2.1 The terms-of-trade gain e from a large-country tariff equals:

- (A) $t \times$ domestic output
- (B) $(P^w - P^{t*}) \times M$: the fall in the price paid to foreign exporters, times imports
- (C) consumer surplus
- (D) the area $b + d$

Q2.2 A small tariff raises a large country's welfare because:

- (A) deadweight losses are always zero
- (B) the terms-of-trade gain e is first-order in t while the distortions $b + d$ are second-order
- (C) foreign producers do not react
- (D) revenue is spent efficiently

Q2.3 The optimum tariff $t^{opt} = 1/\varepsilon_2^*$ implies that against a partner with a very *inelastic* export supply, Home should set:

- (A) a lower tariff
- (B) a higher tariff
- (C) a zero tariff
- (D) an import subsidy

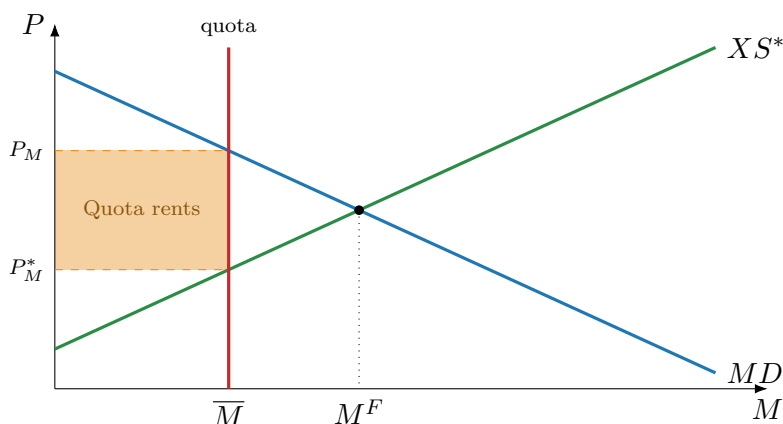
Q2.4 In offer-curve space, the optimal tariff is found where:

- (A) the two offer curves are parallel
- (B) Home's trade indifference curve is tangent to *Foreign's* offer curve
- (C) Foreign's indifference curve is tangent to Home's offer curve
- (D) trade volume is maximized

3 Lesson 3: Import quotas and tariff–quota (non-)equivalence

3.1 Quota rents

An import quota $\bar{M}$ restricts the *quantity* of imports rather than taxing them. With import demand MD and foreign export supply XS^* , a binding quota $\bar{M} < M^F$ raises the domestic price to $P_M = MD^{-1}(\bar{M})$ while the marginal foreign supplier would have been willing to sell at $P_M^* = XS^{*-1}(\bar{M})$: the rectangle $(P_M - P_M^*) \times \bar{M}$ is the **quota rent**. In the figure, $MD : P = 4 - \frac{1}{2}M$, $XS^* : P = \frac{1}{2} + \frac{1}{2}M$; free trade at $M^F = 3.5$, $P = 2.25$; quota $\bar{M} = 2$ gives $P_M = 3$, $P_M^* = 1.5$.



Tariff–quota equivalence and its breakdown

A tariff t and a quota $\bar{M} = M(P^w + t)$ generate the *same* prices and quantities under perfect competition — the only difference is who captures the rectangle: the government (tariff revenue) versus whoever holds the import license (quota rents, f). If rents are captured domestically and redistributed lump-sum, $\Delta W = -b - d + f$, identical to the tariff. If licenses are auctioned to (or captured by) foreigners, $\Delta W = -(b + d)$ —the rectangle: Home loses it entirely. **The equivalence breaks down** once there is domestic market power (Part II): a quota fixes quantity and hands the domestic monopolist full pricing power over the residual demand, which a tariff does not.

Understanding Quiz 3 — Lesson 3

Q3.1 The quota rent is:

- (A) the deadweight loss of the quota
- (B) $(P_M - P_M^*) \times \bar{M}$: the gap between the domestic price and the marginal foreign supply price, on the quota volume
- (C) always captured by the government
- (D) zero when the quota binds

Q3.2 A tariff and an equally restrictive quota differ, under perfect competition, only in:

- (A) the resulting domestic price
- (B) who captures the rectangle: treasury (tariff revenue) vs. license holders (quota rents)
- (C) the volume of imports
- (D) consumer surplus

Q3.3 If import licenses are given to foreign exporters, Home's welfare change from the quota is:

- (A) $-b - d + f$
- (B) $-(b + d)$ minus the entire rent rectangle: strictly worse than the equivalent tariff
- (C) $+f$
- (D) zero

Q3.4 The deep reason tariff-quota equivalence fails under domestic monopoly is:

- (A) quotas are illegal under WTO rules
- (B) a tariff leaves the monopolist facing a price ceiling, while a quota leaves it facing a residual *demand curve* it can exploit
- (C) tariffs raise no revenue
- (D) monopolists never import

Part II

Domestic Market Power and the Theory of the Second Best

When a domestic industry itself has market power, the “first-best” free-trade prescription no longer applies: a well-chosen trade instrument can act as the *second-best* correction for the domestic distortion even absent any terms-of-trade motive. This part develops that case and assembles the game-theory toolkit — best-response functions, strategic substitutes and complements, Cramer's-rule comparative statics, and the envelope theorem — used in every subsequent part.

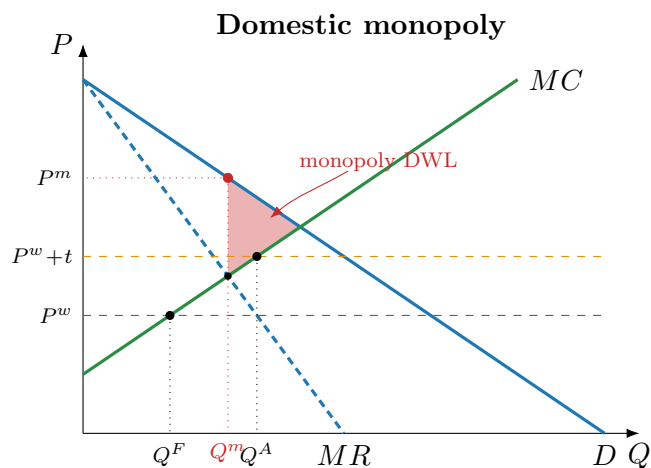
4 Lesson 4: Domestic monopoly protected by a tariff or a quota

4.1 Free trade disciplines the monopolist

Suppose the home industry is a *single* domestic firm facing perfectly elastic foreign supply at P^w . Under free trade the firm cannot price above P^w (it would be undercut by imports), so it behaves as a **price-taker**: $P = P^w$, output Q^F where $MC(Q^F) = P^w$. Import competition completely neutralizes the monopoly distortion.

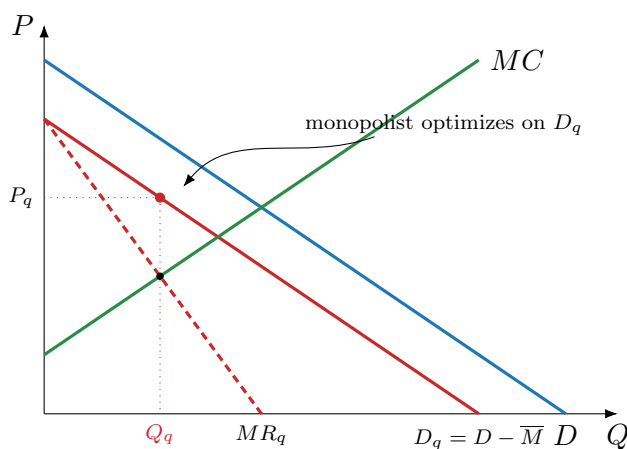
4.2 A tariff vs. a prohibitive tariff

The figure uses $D : P = 6 - Q$, $MR : P = 6 - 2Q$, $MC : P = 1 + Q$, $P^w = 2$, $t = 1$: free-trade domestic output $Q^F = 1$; under the tariff the firm remains a constrained price-taker at $P^w + t = 3$, expanding along MC to $Q = 2$ while consumers demand 3 (imports 1). Only a *prohibitive* tariff makes the firm an unconstrained monopolist: $MR = MC$ at $Q^m = 5/3$, $P^m = 4.33$ — far above the competitive autarky price 3.5 — with the shaded monopoly deadweight loss.



Tariff vs. quota under domestic monopoly

A non-prohibitive tariff keeps the monopolist a price-taker at the ceiling $P^w + t$: welfare falls by the usual $-(b+d)$ but the monopoly distortion stays suppressed. **A quota is worse:** even a *non-prohibitive* quota $\bar{M}$ converts import supply into a fixed quantity, so the monopolist faces the *residual demand* $D_q(P) = D(P) - \bar{M}$ with its own marginal revenue MR_q , and optimizes fully — reintroducing monopoly pricing at *any* binding quota level, however small. This is the deep failure of tariff–quota equivalence flagged in Lesson 3.



Understanding Quiz 4 — Lesson 4

Q4.1 Under free trade, a domestic monopolist facing world price P^w behaves as:

- (A) an unconstrained monopolist
- (B) a price-taker at P^w — import competition removes its pricing power
- (C) a Stackelberg leader
- (D) a perfect price discriminator

Q4.2 A non-prohibitive tariff t leaves the monopolist:

- (A) free to set $MR = MC$
- (B) a constrained price-taker at the higher ceiling $P^w + t$, expanding along MC
- (C) unaffected
- (D) unable to produce

Q4.3 A binding quota reintroduces monopoly pricing because:

- (A) quotas raise marginal cost
- (B) the monopolist faces the downward-sloping residual demand $D(P) - \bar{M}$ rather than a horizontal price ceiling
- (C) quota rents accrue to the government
- (D) consumers stop buying imports

Q4.4 Ranked by welfare (best to worst) for the same import volume under domestic monopoly:

- (A) quota $\succ$ tariff
- (B) tariff $\succ$ quota — the quota adds the monopoly distortion at any binding level
- (C) they are always equivalent
- (D) cannot be ranked

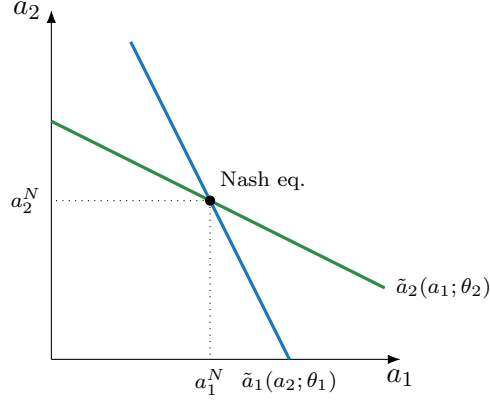
5 Lesson 5: The game-theory toolkit

5.1 Best-response functions and Nash equilibrium

Two players (firms now, governments later) choose actions a_1, a_2 to maximize payoffs $\pi^1(a_1, a_2; \theta_1)$, $\pi^2(a_1, a_2; \theta_2)$, where θ_i is an exogenous shifter (a cost parameter, or a policy of a third party).

First- and second-order conditions

- (1) **Best responses:** $\pi_{a_1}^1 = 0 \Rightarrow \tilde{a}_1(a_2; \theta_1)$, $\pi_{a_2}^2 = 0 \Rightarrow \tilde{a}_2(a_1; \theta_2)$.
- (2) **Concavity and stability:** $\pi_{a_1 a_1}^1 < 0$, $\pi_{a_2 a_2}^2 < 0$, $\Delta \equiv \pi_{a_1 a_1}^1 \pi_{a_2 a_2}^2 - \pi_{a_1 a_2}^1 \pi_{a_2 a_1}^2 > 0$. ($\Delta > 0$ is the Nash stability/uniqueness condition — the analogue of $\Delta = P'(3P' + P''X) > 0$ in the Cournot duopoly of Part III, and of Syropoulos's Marshall–Lerner-type condition $A \equiv \varepsilon^1 + \varepsilon^2 - 1 > 0$ in Part IV.)



5.2 Strategic complements versus substitutes

Definition

$$\pi_{a_1 a_2}^1 \begin{cases} > 0 & \Rightarrow \text{strategic complements (best responses slope up)} \\ < 0 & \Rightarrow \text{strategic substitutes (best responses slope down)} \end{cases}$$

Under strategic *substitutes* (Cournot quantities, tariffs in Part IV) an increase in the rival's action lowers one's own marginal payoff, so one *retreats*. Under strategic *complements* (Bertrand prices, Part V's coda) both actions move together. The sign of this single cross-partial will determine whether optimal export policy is a subsidy or a tax.

5.3 Comparative statics via Cramer's rule

Totally differentiate both first-order conditions with respect to a shock $d\theta_2$ (holding θ_1 fixed):

$$\pi_{a_1 a_1}^1 da_1 + \pi_{a_1 a_2}^1 da_2 = 0, \quad \pi_{a_2 a_1}^2 da_1 + \pi_{a_2 a_2}^2 da_2 + \pi_{a_2 \theta_2}^2 d\theta_2 = 0,$$

$$\begin{pmatrix} da_1 \\ da_2 \end{pmatrix} = \frac{1}{\Delta} \begin{pmatrix} \pi_{a_2 a_2}^2 & -\pi_{a_1 a_2}^1 \\ -\pi_{a_2 a_1}^2 & \pi_{a_1 a_1}^1 \end{pmatrix} \begin{pmatrix} 0 \\ -\pi_{a_2 \theta_2}^2 d\theta_2 \end{pmatrix}.$$

This Cramer's-rule system is the workhorse for *every* comparative-statics result in these notes: it tells us how a shock to one player's parameter ripples through both players' equilibrium actions once strategic feedback is accounted for.

5.4 The envelope theorem

Since player 1 optimizes over a_1 , at the equilibrium $\pi_{a_1}^1 = 0$, so the *direct* effect of a_1 drops out of any welfare comparative static:

$$\frac{d\pi^{1N}}{d\theta_2} = \underbrace{\pi_{a_1}^1}_{=0} \frac{da_1^N}{d\theta_2} + \pi_{a_2}^1 \frac{da_2^N}{d\theta_2}, \quad \frac{d\pi^{2N}}{d\theta_2} = \pi_{a_1}^2 \frac{da_1^N}{d\theta_2} + \underbrace{\pi_{a_2}^2}_{=0} \frac{da_2^N}{d\theta_2} + \pi_{\theta_2}^2.$$

Only effects operating *through the rival's* equilibrium action survive for the player not directly hit — a preview of the profit-shifting logic of Part V. If θ_2 is a tariff-induced cost wedge, $c_2 = c_1 + t$, raising t is formally identical to raising firm 2's marginal cost.

Understanding Quiz 5 — Lesson 5

Q5.1 The stability/uniqueness condition for the Nash equilibrium is:

- (A) $\pi_{a_1}^1 = \pi_{a_2}^2 = 0$
- (B) $\Delta = \pi_{a_1 a_1}^1 \pi_{a_2 a_2}^2 - \pi_{a_1 a_2}^1 \pi_{a_2 a_1}^2 > 0$
- (C) $\pi_{a_1 a_2}^1 > 0$
- (D) $\theta_1 = \theta_2$

Q5.2 Actions are strategic substitutes when:

- (A) $\pi_{a_1 a_2}^1 < 0$: a higher rival action lowers own marginal payoff, so best responses slope down
- (B) $\pi_{a_1 a_2}^1 > 0$
- (C) payoffs are symmetric
- (D) the Nash equilibrium is unique

Q5.3 By the envelope theorem, the effect of a shock θ_2 on player 1's equilibrium payoff runs only through:

- (A) player 1's own action (first-order)
- (B) the rival's equilibrium action, $\pi_{a_2}^1 da_2^N / d\theta_2$
- (C) the constant term
- (D) nothing — payoffs are unaffected

Q5.4 Interpreting a home tariff as $c_2 = c_1 + t$ means:

- (A) tariffs subsidize the foreign firm
- (B) a tariff is formally a rise in the foreign rival's marginal cost of serving the home market
- (C) tariffs change preferences
- (D) the model breaks down

Part III

A Cournot Duopoly Model of Strategic Trade Policy

6 Lesson 6: The duopoly model and cost comparative statics

6.1 Setup

Two firms supply a homogeneous good to the home market: firm 1 (home) with constant marginal cost c_1 , firm 2 (foreign) with constant marginal cost c_2 . Outputs X_1, X_2 , $X = X_1 + X_2$, inverse demand $P(X)$ with $P'(X) < 0$:

$$\pi^1(X_1, X_2) = P(X)X_1 - c_1 X_1, \quad \pi^2(X_1, X_2; c_2) = P(X)X_2 - c_2 X_2.$$

First- and second-order conditions

$$\begin{aligned} \pi_{X_i}^i &= P + P'X_i - c_i = 0, & i = 1, 2, \\ \pi_{X_i X_i}^i &= 2P' + P''X_i < 0 \text{ (SOC)}, & \pi_{X_i X_j}^i = P' + P''X_i < 0 \text{ (strategic substitutes)}, \\ \Delta &\equiv \pi_{X_1 X_1}^1 \pi_{X_2 X_2}^2 - \pi_{X_1 X_2}^1 \pi_{X_2 X_1}^2 > 0. \end{aligned}$$

6.2 Comparative statics with respect to the rival's cost

Totally differentiating both FOCs with respect to c_2 (an increase in the foreign firm's cost, e.g. via a home tariff, $c_2 = c_1 + t$) and applying Cramer's rule:

$$\frac{dX_1}{dc_2} = \frac{-\pi_{X_1 X_2}^1}{\Delta} = \frac{-(P' + P''X_1)}{\Delta} > 0, \quad \frac{dX_2}{dc_2} = \frac{2P' + P''X_1}{\Delta} < 0, \quad \frac{dX}{dc_2} = \frac{P'}{\Delta} < 0.$$

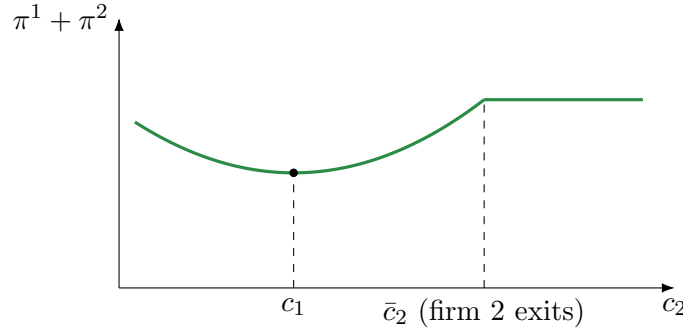
An increase in the rival's cost raises own output, lowers the rival's output by more, and lowers total output (raising price) — the standard Cournot cost pass-through, and the reason a tariff on a foreign duopolist can *shift profit* toward the home firm.

6.3 Effects on profits

Using $\pi_{X_2}^1 = P'X_1 < 0$, $\pi_{c_2}^2 = -X_2 < 0$, $\pi_{X_1}^2 = P'X_2 < 0$ and the envelope theorem:

$$\frac{d\pi^1}{dc_2} = \pi_{X_2}^1 \frac{dX_2}{dc_2} = (-)(-) > 0, \quad \frac{d\pi^2}{dc_2} = \pi_{X_1}^2 \frac{dX_1}{dc_2} + \pi_{c_2}^2 = (-)(+) + (-) < 0.$$

Own profit always rises, the rival's always falls — but the effect on the *sum* $\pi^1 + \pi^2$ is ambiguous, producing the U-shape below: near $c_2 = c_1$, business-stealing dominates and the joint pie shrinks; for larger wedges the home-firm gain dominates; once firm 2 is priced out ($X_2 \rightarrow 0$ at $\bar{c}_2$), joint profit flattens at the home-monopoly level.



6.4 The symmetric benchmark

At the symmetric point $c_2 = c_1$, $X_1 = X_2 = X$:

$$\Delta = P'(3P' + P''X) > 0, \quad \left. \frac{d(\pi^1 + \pi^2)}{dX_2} \right|_{c_2=c_1} = X_1 \left[\frac{(P')^2}{\Delta} - 1 \right] < 0 :$$

starting from symmetry, a small unilateral increase in firm 2's cost lowers *joint industry profit* — but this is not the same as national welfare, which also counts consumer surplus.

Understanding Quiz 6 — Lesson 6

Q6.1 In Cournot with $P' + P''X_i < 0$, quantities are:

- (A) strategic complements
- (B) strategic substitutes: reaction functions slope down
- (C) independent
- (D) perfect substitutes in preferences

Q6.2 A rise in the foreign rival's cost c_2 causes:

- (A) $X_1 \downarrow, X_2 \uparrow$
- (B) $X_1 \uparrow, X_2 \downarrow$, total output $\downarrow$, price $\uparrow$
- (C) both outputs rise
- (D) no change in price

Q6.3 The effect of c_2 on joint profit $\pi^1 + \pi^2$ is:

- (A) always positive
- (B) always negative
- (C) U-shaped: negative near symmetry, positive for large wedges, flat after exit
- (D) exactly zero

Q6.4 Why does own profit *always* rise when the rival's cost rises?

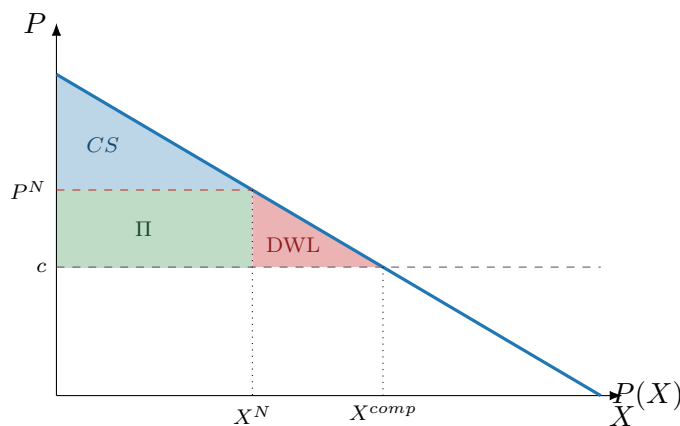
- (A) demand shifts out
- (B) by the envelope theorem the only surviving channel is the rival's contraction, which raises the price firm 1 receives
- (C) costs fall for firm 1
- (D) consumers switch loyalty

7 Lesson 7: Welfare, the marginal tariff, and reciprocal dumping

7.1 Quasilinear national welfare

Add a numeraire good Z (price 1, produced with labor alone at wage $w = 1$) and quasilinear preferences $U(X, Z) = u(X) + Z$ with budget $PX + Z = wL + \pi^1 + \pi^2$. Then

$$U = \underbrace{u(X) - PX}_{CS} + \underbrace{\pi^1 + \pi^2}_{\Pi} + wL, \quad u'(X) = P \text{ (inverse demand = marginal utility).}$$



Cournot market power already creates a deadweight loss at free trade ($P^N > c$): this pre-existing distortion is exactly why trade policy operates in a *second-best* world here.

7.2 The marginal tariff at free trade

Let $c_2 = c_1 + t$. Differentiating $U = u(X) - PX + \pi^1 + \pi^2$ with $u'(X) = P$ and the envelope theorem on each firm's optimized profit:

$$dU = (P - c_1) dX_1 + (P - c_2) dX_2 - X_2 dt = (P - c_2) dX + t dX_1 - X_2 dt.$$

Result: the sign of the marginal tariff

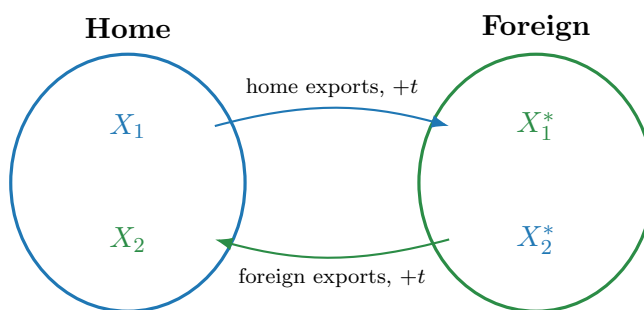
Evaluated at $t = 0$:

$$dU|_{t=0} = -X_2 dt < 0.$$

A marginal home tariff, starting from free trade, is welfare-reducing in this baseline single-market Cournot model: the profit-shifting gain is second-order (an envelope effect), while the consumption loss from the higher price is first-order. A *first-order* gain from a tariff requires either (i) genuine terms-of-trade power in a competitive model (Lesson 2), or (ii) a strictly positive profit-shifting motive of the Brander–Spencer type (Part V) — and even then, worked examples show the *optimal* tariff can be strictly positive once profit-shifting operates globally rather than at the margin.

7.3 Reciprocal dumping: the two-country extension

Suppose *both* countries host one firm and each ships to both markets (“cross-hauling”): the home firm sells X_1 at home and X_2^* abroad; the foreign firm sells X_2 at home and X_1^* abroad; each export leg bears the trade cost/tariff t .



By symmetry $X_1^* = X_2$ and $X_2^* = X_1$, and Home welfare adds the home firm’s *foreign-market* profit:

$$U = \underbrace{u(X_1 + X_2) - P(X_1 + X_2)}_{CS} + \pi^1 + \pi^{1*}(X_1^*, X_2^*, t) + wL.$$

This intra-industry-trade structure (Brander 1981; Brander–Krugman) is the bridge to national welfare analysis of strategic trade policy and to the two-government games of Parts IV–V. A useful building block: with quasilinear welfare $U = u(Q) - PQ$, $dU = (u'(Q) - P)dQ + Q dP = Q dP$ at the consumer optimum, and along the demand curve $dQ = dP/P'(Q)$. Under the regularity condition $P' + P''x < 0$, a firm’s exports and the price it faces move *together*: $dx > 0 \iff dP > 0$ — the sign tool used repeatedly in Part V.

Understanding Quiz 7 — Lesson 7

Q7.1 In the quasilinear framework, national welfare is:

- (A) consumer surplus only
- (B) $CS + \text{total profits (+ wage income)}$: $u(X) - PX + \pi^1 + \pi^2 + wL$
- (C) tariff revenue only
- (D) GDP at world prices

Q7.2 At free trade ($t = 0$), a marginal home tariff changes welfare by:

- (A) $+X_2 dt$

- (B) $-X_2 dt < 0$: the first-order consumer loss dominates the second-order profit-shifting gain
- (C) zero
- (D) $+t dX_1$

Q7.3 “Reciprocal dumping” refers to:

- (A) both governments subsidizing agriculture
- (B) two-way trade in the *same* homogeneous good, driven by Cournot rivalry across segmented markets
- (C) selling below cost in one’s own market
- (D) exchange-rate manipulation

Q7.4 Cournot market power matters for trade-policy analysis because:

- (A) it eliminates deadweight loss
- (B) it creates a pre-existing distortion ($P > c$), putting policy in a second-best world where interventions can raise welfare
- (C) it makes demand vertical
- (D) it fixes the terms of trade

Part IV

Optimum Tariffs, Retaliation, and Country Size: Three Classics

This part connects the reaction-function machinery of Part II to the canonical theory of strategic tariff policy: Johnson (1953–54), Mayer (1981), Syropoulos (2002).

8 Lesson 8: Johnson (1953–54) and Mayer (1981)

Johnson, H.G. (1953–54), *Review of Economic Studies*

Setting. Two-country, two-good exchange model with community indifference curves and offer curves O_I, O_{II} . **Key results.**

- Country I’s optimal unilateral tariff is $t_I = \frac{1}{j-1}$, where j is the elasticity of Country II’s offer curve at the post-tariff equilibrium — the $t^{opt} = 1/\varepsilon^*$ formula of Lesson 2.
- Because tariff proceeds are redistributed to consumers, the tariff *rotates* Home’s offer curve (it does not shift it proportionally).
- **Retaliation:** Country II’s best response is its own optimum tariff $t_{II} = 1/(k-1)$. Iterating (a reaction-curve technique Johnson explicitly likens to the *Cournot duopoly solution*) converges either to a **Nash policy equilibrium** or to a perpetual **tariff cycle**.
- **Central result (contra Scitovsky):** it is *not* true that both countries must lose from retaliation relative to free trade. At most one country must lose (mutual terms-of-trade improvement is impossible), but either country *can* end up better off under mutual

tariffs, depending on the relative elasticities j, k of the two offer curves (constant-elasticity example $X = AY^k$, $Y = BX^j$); trade volume never falls to zero.

- Who “wins” the tariff war is governed entirely by relative reciprocal-demand elasticities — there is no simple universally rankable index (Johnson’s criticism of Kaldor).

Mayer, W. (1981), *Oxford Economic Papers*

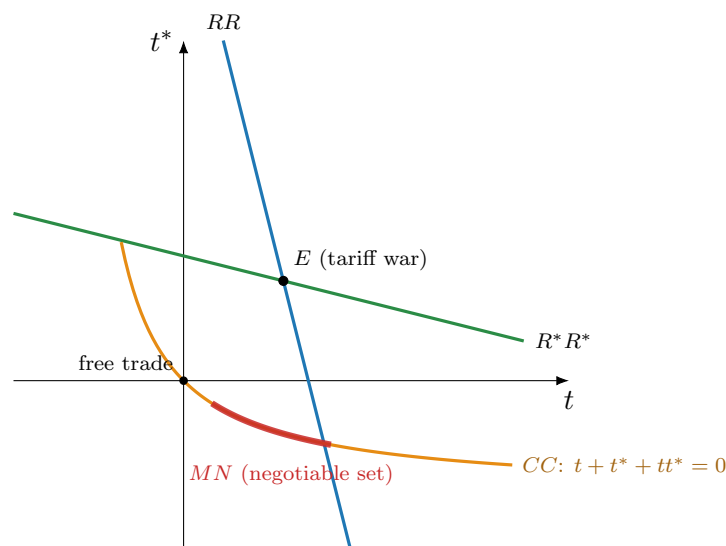
Setting. Two large countries with welfare functions $W(t, t^*)$, $W^*(t, t^*)$ defined directly over the two tariffs. **Key constructs.**

- The *RR locus*: Home’s best-response tariff to every foreign tariff (the zero-slope points of Home’s tariff-indifference curves); similarly R^*R^* . Their intersection E is the **tariff-war (Nash) equilibrium** — the threat point of any negotiation.
- The *CC locus*: the Pareto-efficient tangency locus of the two countries’ tariff-indifference curves. Along CC domestic relative prices are equalized across countries, which (using $p = \pi(1 + t)$, $p^* = \pi/(1 + t^*)$) implies

$$t + t^* + t t^* = 0$$

— **tariffs by both countries cannot be Pareto efficient**: either free trade for both, or a tariff by one paired with a *subsidy* by the other.

- **Negotiated settlements**: the feasible agreements form the segment $MN \subset CC$ that Pareto-dominates E . Since E features *perpendicular* indifference curves while CC requires *tangency*, some point of CC strictly dominates E : **negotiated tariffs always (weakly) beat a tariff war**.
- **Free trade need not be attainable**: $(0, 0)$ lies on CC but not necessarily on MN ; bargaining from the tariff-war threat point can settle on an efficient outcome that still involves a tariff–subsidy pair. In the large–small case the small country’s threat tariff is already zero, further restricting the negotiable range.



The figure (drawn with linear best responses and the exact locus $t^* = -t/(1+t)$) shows Mayer's architecture: the tariff war lands at E , deep inside the inefficient region; every efficient point lies on CC , where one instrument is a *subsidy*; the negotiation settles somewhere on the highlighted MN segment — which may or may not include free trade $(0,0)$.

Understanding Quiz 8 — Lesson 8

Q8.1 Johnson's central finding on retaliation is that:

- (A) both countries always lose a tariff war relative to free trade
- (B) at most one country must lose; a country with favorable relative offer-curve elasticities can *win* the tariff war
- (C) trade always collapses to zero
- (D) tariffs converge to 100%

Q8.2 Mayer's Pareto-efficiency condition $t + t^* + tt^* = 0$ implies:

- (A) both countries should set equal tariffs
- (B) efficiency requires free trade for both, or a tariff by one *matched by a subsidy* by the other
- (C) tariffs are efficient if small
- (D) only quotas are efficient

Q8.3 The threat point of Mayer's negotiation is:

- (A) free trade
- (B) the Nash tariff-war equilibrium E at the crossing of RR and R^*R^*
- (C) autarky
- (D) the point M

Q8.4 Negotiated settlements always dominate the tariff war because:

- (A) governments dislike conflict
- (B) at E the indifference curves cross (perpendicular slopes), so tangency points on CC exist that make both better off
- (C) the WTO enforces free trade
- (D) E lies on CC

9 Lesson 9: Syropoulos (2002) — how country size matters

Syropoulos, C. (2002), *Review of Economic Studies*, 69, 707–727

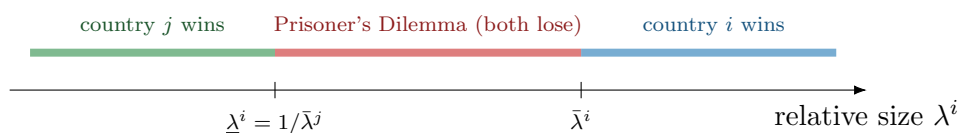
Setting. General neoclassical two-country, two-good model (GDP/expenditure functions à la Dixit–Norman); country i 's **relative size** is $\lambda^i \equiv L^j/L^i$, holding per-capita endowments fixed — cleaner than GDP shares because it is exogenous to trade policy. **Key results.**

- Nash tariffs satisfy Johnson's formula in ad valorem, general-equilibrium form:

$$\tau^i = \frac{1}{\varepsilon^i - 1}, \quad \varepsilon^i = \text{partner's import-demand elasticity,}$$

generating tariff best responses $\tau^1(\tau^2)$, $\tau^2(\tau^1)$ with interior Nash equilibrium N .

- Proposition 1 (limits).** As a partner becomes infinitesimally small ($\lambda^j \rightarrow 0$), its prices, welfare, and tariff converge to its own free-trade/autarky levels, while the large country's domestic price converges to its own autarky price under both free trade and retaliation.
- Proposition 2 (big countries can win).** For each i there is a threshold relative size $\bar{\lambda}^i$ such that i wins the tariff war (better off under retaliation than under free trade) whenever $\lambda^i > \bar{\lambda}^i$. With $\underline{\lambda}^i \equiv 1/\bar{\lambda}^j$, sizes split into three regions: j wins, a **Prisoner's Dilemma** region (both lose), and i wins.
- Assumptions 1–2** ($\theta^i \equiv p^i \varepsilon_{p^i}^i > 0$; substitution elasticity $\sigma^i > 1$) imply **Proposition 3**: tariffs are **strategic substitutes**, the Nash equilibrium is **unique**, and each country's tariff is **increasing in own size** and **decreasing in the partner's** — the tariff-space analogue of $dX_1/dc_2 > 0 > dX_2/dc_2$ from Lesson 6.
- Proposition 4**: the size threshold is unique and the Prisoner's Dilemma zone is exactly $\underline{\lambda}^i < \lambda^i < \bar{\lambda}^i$ with $\underline{\lambda}^i = 1/\bar{\lambda}^j < 1$.
- A **Heckscher–Ohlin** special case makes the size $\rightarrow$ elasticity $\rightarrow$ tariff channel transparent, illustrating the convergence in Proposition 1.



9.1 Synthesis: three papers, one force

	Johnson (1953–54)	Mayer (1981)	Syropoulos (2002)
Model	2-country exchange, offer curves	Tariff-space negotiation	General GE, size λ^i
Optimal tariff	$t = 1/(j - 1)$	Same, via RR locus	$\tau^i = 1/(\varepsilon^i - 1)$
Retaliation	Cournot-style reaction curves; cycles possible	Threat point E (Nash)	Unique Nash under A1–A2
Who wins?	Relative elasticities j, k	— (focus on negotiation)	Relative size vs. threshold
Cooperation	Not modeled	CC dominates E ; $t+t^*+tt^* = 0$	Not modeled

In the Cournot model the “elasticity” that determines who gains from a cost/tariff wedge is the curvature of demand; in Johnson and Syropoulos it is the partner’s offer-curve/import-demand elasticity. In every case the same force operates: **the side with the less elastic response bears more of the burden, and the intervening party captures the difference** — subject to Mayer’s caveat that unilateral optimization is always dominated by a negotiated, Pareto-efficient settlement.

Understanding Quiz 9 — Lesson 9

Q9.1 Syropoulos measures country size by relative population $\lambda^i = L^j/L^i$ (at fixed per-capita endowments) because:

- (A) population data are more accurate
- (B) GDP shares are endogenous to trade policy, while λ^i is a true primitive
- (C) small countries have no GDP
- (D) it makes the algebra harder

Q9.2 Proposition 2 states that a country wins the tariff war if:

- (A) it moves first
- (B) its relative size exceeds a threshold $\bar{\lambda}^i$
- (C) it has absolute advantage in both goods
- (D) its tariff is lower

Q9.3 Under Assumptions 1–2, tariffs are:

- (A) strategic complements
- (B) strategic substitutes, with a unique Nash equilibrium; own tariff rises with own size
- (C) neutral
- (D) random

Q9.4 The “Prisoner’s Dilemma” region is where:

- (A) both countries win the tariff war
- (B) both countries lose relative to free trade, yet each finds a positive tariff unilaterally optimal
- (C) neither sets a tariff
- (D) only the small country loses

Part V

Strategic Trade and Industrial Policy under Imperfect Competition

This part completes the roadmap: the *foreign-monopoly* tariff problem with general demand, the tariff as profit-shifting device, the Brander–Spencer theory of strategic *export* subsidies, and the price-versus-quantity caveat that governs whether the optimal export policy is a subsidy or a tax.

10 Lesson 10: Import policy — rent extraction and profit shifting

10.1 Foreign monopoly and the tariff

A single home market is served by a foreign monopolist exporting X at marginal cost c , facing a home tariff t . National welfare is $U = u(X) - PX + tX$ (no domestic producer yet). The monopolist:

$$\pi(X) = PX - (c + t)X, \quad \text{FOC: } P + P'X - (c + t) = 0, \quad \text{SOC: } 2P' + P''X < 0.$$

Totally differentiating the FOC: $X_t = \frac{dX}{dt} = \frac{1}{2P' + P''X} < 0$. Welfare maximization ($U_t = 0$, using $u' = P$ and the envelope theorem) gives

$$t^{opt} = X \left(P' - \frac{1}{X_t} \right) = X (P' - (2P' + P''X)) = -X (P' + P''X).$$

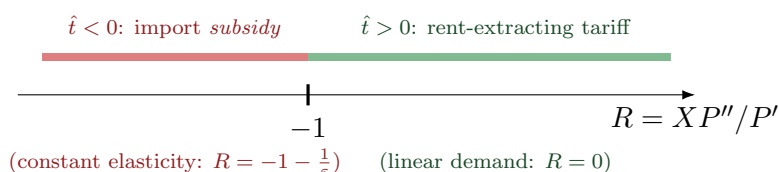
Brander and Spencer (1984): the sign of the optimal tariff

Define the **relative convexity of demand** $R \equiv XP''/P'$. Then $\hat{t} = -P'X(R + 1)$ and:

Proposition. $\hat{t} \gtrless 0$ as $R \gtrless -1$.

- *Linear demand* ($P'' = 0 \Rightarrow R = 0$): $\hat{t} = -P'X > 0$ — the familiar positive **rent-extraction** tariff on a foreign monopolist.
- *Constant-elasticity demand* $P = X^{-1/\varepsilon}$: direct computation gives $R = -1 - 1/\varepsilon < -1$ for every $\varepsilon > 0$, so $\hat{t} < 0$ **always**: the optimal policy is an *import subsidy*! With highly convex demand the tariff passes through *more* than one-for-one into the domestic price (pass-through $dP/dt = 1/(R + 2) > 1$ when $R < -1$), so the consumer cost of extracting rent swamps the revenue gained.

A cautionary result: “tax the foreign monopolist” depends delicately on demand curvature, not merely on the existence of market power.



10.2 Domestic and foreign firms: profit shifting

Reintroduce a domestic firm with output y ($\pi(y) = yP(X) - cy - F$) alongside foreign imports x ($V(x, y; t) = xP(X) - kx - tx$, $X = x + y$), Cournot competition, revenue redistributed lump-sum:

$$G(t) = u(X) - PX + tX + \pi(y).$$

Stability requires $D \equiv \pi_{yy}V_{xx} - \pi_{yx}V_{xy} > 0$; comparative statics give

$$y_t > 0, \quad x_t < 0, \quad X_t = y_t + x_t < 0.$$

Proposition (Brander–Spencer 1984): the tariff shrinks consumption but can still pay

Total consumption always falls ($X_t < 0$), yet the country typically gains: the tariff shifts output and profit from the foreign to the domestic firm and collects revenue on residual imports. Setting $G_t = 0$ nets the consumption loss against the profit-shifting/revenue gain; in the linear worked example the optimum is strictly positive ($t^{opt} = 2\alpha/11$) — profit shifting eventually dominates the consumption loss even though the *marginal* tariff at free trade was welfare-reducing in Lesson 7’s one-firm-each benchmark.

Two countries: transport costs and jointly excessive protection

In the two-country reciprocal-dumping extension (each country one firm, iceberg cost k): **(i)** a higher k *lowers* each country’s optimal tariff — transport costs already choke off intra-industry trade, leaving less rent to extract or shift; **(ii)** at the noncooperative equilibrium, *world* welfare would rise if both tariffs were cut below their unilateral optima: noncooperative tariffs are **jointly excessive** — the same message as Mayer’s dominance of negotiation over the tariff war (Lesson 8).

Understanding Quiz 10 — Lesson 10

Q10.1 Against a foreign monopolist with *linear* demand, the optimal home policy is:

- (A) free trade
- (B) a positive rent-extracting tariff $\hat{t} = -P'X > 0$
- (C) an import subsidy
- (D) a quota of zero

Q10.2 With constant-elasticity demand, the optimal policy flips to an import subsidy because:

- (A) elasticity is infinite
- (B) $R = -1 - 1/\varepsilon < -1$: tariff pass-through exceeds one-for-one, so consumers lose more than the treasury gains
- (C) the monopolist has no rents
- (D) revenue cannot be collected

Q10.3 In the domestic-plus-foreign-firm model, a tariff raises home welfare mainly by:

- (A) raising total consumption
- (B) shifting output and profit from the foreign to the domestic firm ($y_t > 0$, $x_t < 0$) plus revenue on residual imports
- (C) lowering the domestic price

(D) eliminating fixed costs

Q10.4 Higher transport costs k lower optimal tariffs because:

- (A) governments cannot measure them
- (B) markets are already more segmented, so less rent remains for a tariff to extract or shift
- (C) they are a substitute form of revenue
- (D) firms merge

11 Lesson 11: Export policy — Brander–Spencer (1985) and the Bertrand caveat

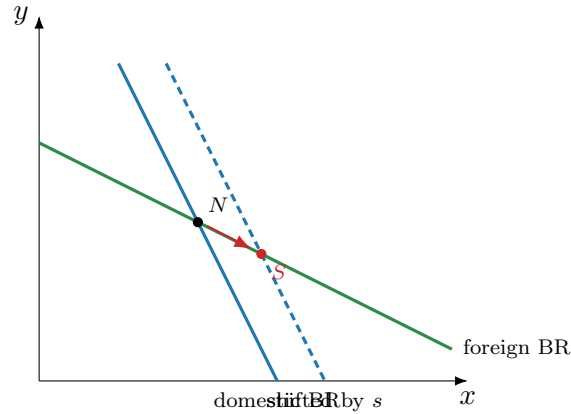
11.1 The third-market model

A domestic firm (output x) and a foreign firm (output y) compete Cournot in a *third*, purely importing market. With a per-unit export subsidy s :

$$\pi(x, y; s) = xP(x + y) - c(x) + sx, \quad \pi_x = xP' + P - c_x + s = 0, \quad \pi_{xx} = 2P' + xP'' - c_{xx} < 0,$$

and symmetrically $\pi_y^* = yP' + P - c_y^* = 0$ for the foreign firm. Regularity: $\pi_{xy} = P' + xP'' < 0$, $\pi_{yx}^* = P' + yP'' < 0$ (downward reaction functions), $D \equiv \pi_{xx}\pi_{yy}^* - \pi_{xy}\pi_{yx}^* > 0$. Totally differentiating with respect to s ($\pi_{xs} = 1$, $\pi_{ys}^* = 0$):

$$x_s = \frac{-\pi_{yy}^*}{D} > 0, \quad y_s = \frac{\pi_{yx}^*}{D} < 0.$$



Proposition 1 (Brander–Spencer 1985)

An increase in the domestic export subsidy: (i) lowers the world price, $p_s = P'(x_s + y_s) < 0$; (ii) raises domestic profit, $\pi_s = xP'y_s + x > 0$ (envelope theorem: $\pi_s = x$ directly, plus the gain from the rival's retreat); (iii) lowers foreign profit, $\pi_s^* = yP'x_s < 0$.

11.2 Optimal subsidy and the Stackelberg interpretation

With all output exported, domestic welfare is $G(s) = \pi - sx$. Using $\pi_s = x$:

$$G_s = xP'y_s - sx_s \quad \implies \quad G_s|_{s=0} = xP'y_s > 0, \quad \boxed{s^{opt} = \frac{xP'y_s}{x_s} > 0.}$$

A marginal export subsidy always raises welfare from free trade — the mirror image of Lesson 7’s tariff result, with the opposite sign because there is no domestic consumer loss to offset the profit shift.

Proposition 3: the subsidy replicates Stackelberg leadership

The optimal subsidy moves the equilibrium to exactly the point the domestic firm would choose if it were a **Stackelberg quantity leader** without any subsidy. *Sketch:* the leader’s FOC is $\pi_x + \pi_y(dy/dx) = 0$ with $dy/dx = y_s/x_s$ the slope of the foreign reaction function; substituting $s^{opt} = xP'y_s/x_s$ into the subsidized Cournot FOC $\pi_x(x, y; s) = 0$ reproduces it exactly. Diagrammatically, the subsidy shifts the domestic reaction function out, moving the equilibrium from N to S : more domestic output, less foreign output, higher domestic profit net of the subsidy.

Rivalry is essential: for an international *monopolist* (no rival), $\pi_s = x$ still, but $G_s = -s x_s < 0$ for any $s > 0$ — with no one to steal profit from, the subsidy is pure waste.

11.3 General equilibrium, subsidy wars, and three governments

In a two-good GE with quasilinear utility and home consumption z , the change in real income is $d\mathcal{I} = (x - z)dp + (p - c_x)dx$: under perfect competition ($p = c_x$) any subsidy only worsens the terms of trade — which is why export subsidies make no sense in a competitive neoclassical world. Under Cournot rivalry $p - c_x = -xP' - s > 0$, and $s^{opt} = \frac{xP'y_s}{x_s} - \frac{zp_s}{x_s} > 0$ survives.

Propositions 4–5: the subsidy war is a Prisoner’s Dilemma

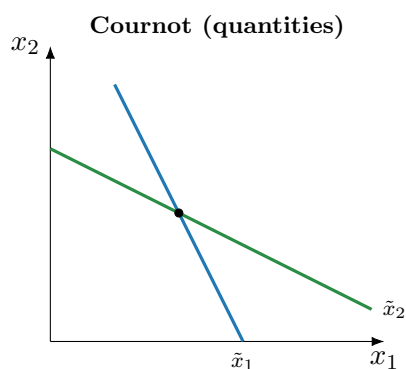
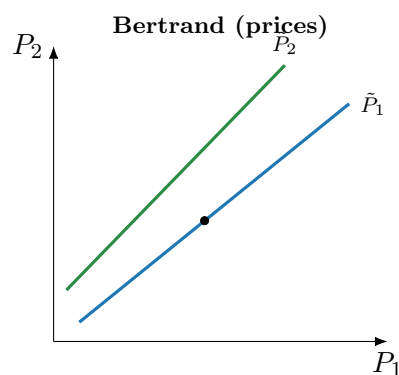
If both exporting countries subsidize, the Nash equilibrium has *both* subsidies positive — yet joint exporter welfare would rise if both were cut below Nash levels. If the exporters consumed none of the good, the *jointly* optimal policy would be an export **tax** (a two-firm cartel against the importer) — but it is not self-enforcing. Adding the importing country’s own optimal tariff t closes the triangle: with linear demand and no home consumption,

$$s = -\frac{1}{2}xP', \quad s^* = -\frac{1}{2}yP', \quad t = -\frac{1}{2}XP' = s + s^* :$$

the importer’s optimal tariff *equals the sum* of the exporters’ optimal subsidies; each subsidy raises the optimal tariff (the importer taxes back part of the subsidy-induced rents), and a higher tariff lowers each optimal subsidy.

11.4 The cautionary coda: Cournot vs. Bertrand

Everything above assumed *quantity* competition, making reaction functions downward-sloping (**strategic substitutes**) so that committing to aggression (a subsidy) induces the rival to *re-treat*. Under *price* competition for differentiated products, reaction functions slope *up* (**strategic complements**): aggression triggers matching aggression.

strategic substitutes: BRs slope *down*strategic complements: BRs slope *up*

Why the sign of optimal export policy flips

Under Bertrand, an export **tax** raises the domestic firm's effective cost and optimal price, shifting its price-reaction curve up; the rival then *also raises* its price, softening competition and raising both margins — including the domestic firm's, net of the tax. A subsidy would trigger a rent-dissipating price war. Whether profit-shifting calls for a **subsidy** (Cournot) or a **tax** (Bertrand) hinges entirely on the sign of $\pi_{a_1 a_2}$ from Lesson 5 — a sharp warning against applying the Brander–Spencer subsidy prescription without verifying how firms actually compete.

Understanding Quiz 11 — Lesson 11

Q11.1 At free trade, a marginal export subsidy in the third-market Cournot model:

- (A) lowers welfare, like the marginal tariff of Lesson 7
- (B) raises welfare: $G_s|_{s=0} = xP'y_s > 0$, with no domestic consumer loss to offset the profit shift
- (C) has no effect
- (D) raises the world price

Q11.2 The optimal subsidy implements the outcome the home firm would achieve as:

- (A) a perfect competitor
- (B) a Stackelberg quantity leader
- (C) a monopolist
- (D) a Bertrand player

Q11.3 For an international monopolist with no rival, an export subsidy is:

- (A) still optimal
- (B) pure waste: $G_s = -s x_s < 0$ — there is no rival from whom to shift profit
- (C) equivalent to a tariff
- (D) necessary for exports

Q11.4 Under Bertrand competition with differentiated products, the optimal export policy is typically:

- (A) a larger subsidy
- (B) an export *tax*: with upward-sloping reaction curves, committing to softness raises both prices and margins
- (C) free trade always
- (D) a quota

Q11.5 In the three-government linear benchmark, the importer's optimal tariff equals:

- (A) zero
- (B) the sum of the two exporters' optimal subsidies, $t = s + s^*$
- (C) half of each subsidy
- (D) the world price

Part VI

Rent-Seeking and the Political Economy of Trade Policy

Parts I–V derived trade policy from a benevolent, welfare-maximizing government. This part asks the opposite question: if governments respond to *political* pressure — voters, lobbies, campaign contributions — what tariff structure emerges, and how does it compare to the welfare-maximizing benchmarks? Two complementary answers: Mayer (1984), where tariffs emerge from **majority voting** by factor owners; and Grossman–Helpman (1994), where tariffs emerge from **lobbying and campaign contributions** in a common-agency game.

12 Lesson 12: Mayer (1984) — endogenous tariffs by majority voting

12.1 Heterogeneous factor ownership

A small open Heckscher–Ohlin economy (goods 1, 2; capital and labor) imports good 1. Every individual i owns one unit of labor and $K^i \geq 0$ units of capital:

$$y^i = w + rK^i + \theta^i T, \quad \theta^i \equiv \frac{w + rK^i}{wL + rK},$$

with tariff revenue T redistributed in proportion to factor-income shares. Indirect utility is $U^i(p, y^i)$ with $p = \pi(1 + t)$.

Individually optimal tariffs

Differentiating (Roy's identity, homothetic preferences):

$$\frac{dU^i}{dt} = \frac{\partial U^i}{\partial y^i} \left[\theta^i \pi \frac{dM}{dt} + Y \frac{d\theta^i}{dt} \right].$$

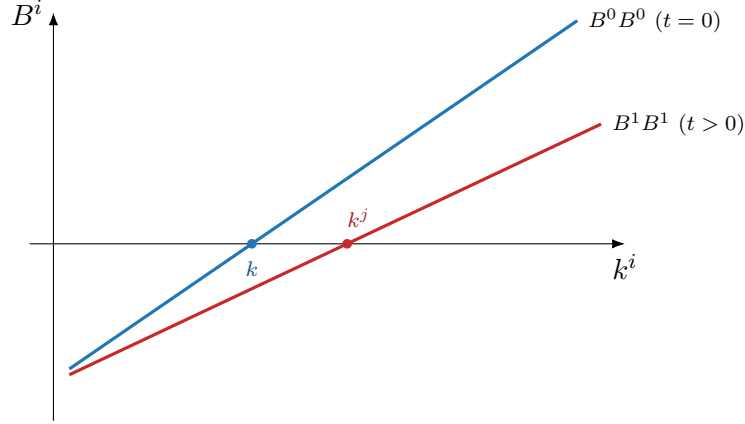
Since $dM/dt < 0$ always, person i 's optimal tariff $\hat{t}^i$ is positive, zero, or negative according to whether a higher tariff *raises* i 's income share θ^i — which happens iff i 's own capital–labor ratio k^i exceeds the national ratio k and the import good is capital-intensive (mirror-image otherwise). In short:

$$\hat{t}^i \geq 0 \iff \text{person } i \text{ is relatively well (poorly) endowed with the import good's intensive factor.}$$

A person with $k^i = k$ is indifferent ($\hat{t}^i = 0$); the farther k^i from k , the larger $|\hat{t}^i|$.

12.2 The median-voter tariff

Each voter's welfare is single-peaked in t , so Black's median-voter theorem applies: majority voting selects the **median** voter's optimal tariff, $t^* = t^m$. Comparing median to mean: capital ownership is empirically right-skewed (many small owners, few large), so the median capital ratio lies *below* the mean — generating a built-in political bias toward protecting the *labor*-owning majority (e.g. tariffs on capital-intensive imports), unless voting eligibility or participation offsets it.



The locus $B^i(k^i, t)$ — the real-income gain from a marginal tariff increase — is upward sloping in k^i (more capital, more to gain from protecting the capital-intensive sector) and shifts *down* as t rises (strict concavity: further protection helps fewer people). The zero-crossing at $t = 0$ is the national ratio k ; at a positive tariff it moves right to k^j : person j with $k^j > k$ is exactly the voter whose optimum is that tariff.

12.3 Voting costs, participation, and the specific-factors asymmetry

With costly voting, only those whose stake $|B^i|$ exceeds the participation cost vote; equilibrium equates votes for and against, and the tariff depends on the *shape of both tails* of the ownership distribution, not just the median. Mayer's most quoted result then comes from the multi-sector *specific-factors* version: a tariff on good g raises the income of g 's specific-factor owners by

$$B_{gg} = Y\beta_g(1-\beta_g)\chi_g^i > 0, \quad \text{while hurting each other sector's owners by only} \quad B_{hg} = -Y\beta_g\beta_h\chi_h^i < 0,$$

where β_g is g 's share of national income.

Concentrated benefits, diffuse costs

With more than two sectors, $(1-\beta_g) > \beta_h$ for every $h \neq g$, so $B_{gg} > -B_{hg}$ for equal ownership shares: **per capita, gains to the protected sector's owners always exceed the losses to any other single sector's owners.** With participation costs, a small, well-organized sector can secure protection despite being an electoral minority: numerous losers each lose little (and stay home), while concentrated gainers organize and vote — a rigorous version of Olson's logic of collective action, and the conceptual bridge to Grossman–Helpman.

Understanding Quiz 12 — Lesson 12

Q12.1 In Mayer (1984), person i wants a positive tariff iff:

- (A) they consume no imports
- (B) they are relatively well endowed with the factor used intensively in the import-competing sector
- (C) their income is above average
- (D) they own no capital

Q12.2 With costless universal voting, the equilibrium tariff is:

- (A) the mean voter's optimum
- (B) the median voter's optimal tariff (single-peaked preferences, Black's theorem)
- (C) zero
- (D) the maximum-revenue tariff

Q12.3 Right-skewed capital ownership biases the vote toward:

- (A) free trade
- (B) policies favoring the labor-owning majority, e.g. tariffs on capital-intensive imports
- (C) export subsidies
- (D) capital owners

Q12.4 The specific-factors result $B_{gg} > -B_{hg}$ explains why:

- (A) all sectors are protected equally
- (B) small, organized sectors win protection: concentrated per-capita gains beat diffuse per-capita losses once participation is costly
- (C) tariffs are zero in equilibrium
- (D) consumers organize easily

13 Lesson 13: Grossman–Helpman (1994) — Protection for Sale

13.1 The common-agency model

A small competitive economy has quasilinear preferences ($V(p, E) = E + s(p)$, $s(p)$ consumer surplus). Good 0 is the labor-only numeraire; each other good uses labor plus a sector-specific factor earning $\pi_i(p_i)$. Trade taxes/subsidies drive wedges $t_i = (p_i - p_i^*)/p_i^*$; revenue $r(p)$ is redistributed uniformly. Aggregate welfare: $W(p) = \sum_i \pi_i(p_i) + N[r(p) + s(p)]$. In an exogenous subset L of sectors, specific-factor owners are organized into **lobbies**. Lobby i offers a **contribution schedule** $C_i(p)$; the government maximizes

$$G = \sum_{i \in L} C_i(p) + a W(p), \quad a \geq 0$$

— a political-support function that is *derived*, not assumed.

13.2 Truthful equilibria and the tariff formula

This is a **common agency / menu auction** (Bernheim–Whinston 1986): the relevant (coalition-proof) equilibria are supported by **truthful** schedules $C_i^T(p, B_i) = \max[0, W_i(p) - B_i]$, and the equilibrium price vector solves

$$p^0 = \arg \max_p \left[\sum_{i \in L} W_i(p) + a W(p) \right] :$$

the government acts *as if* maximizing a welfare function weighting lobby members $1 + a$ and everyone else a .

Proposition 2 (Grossman–Helpman 1994)

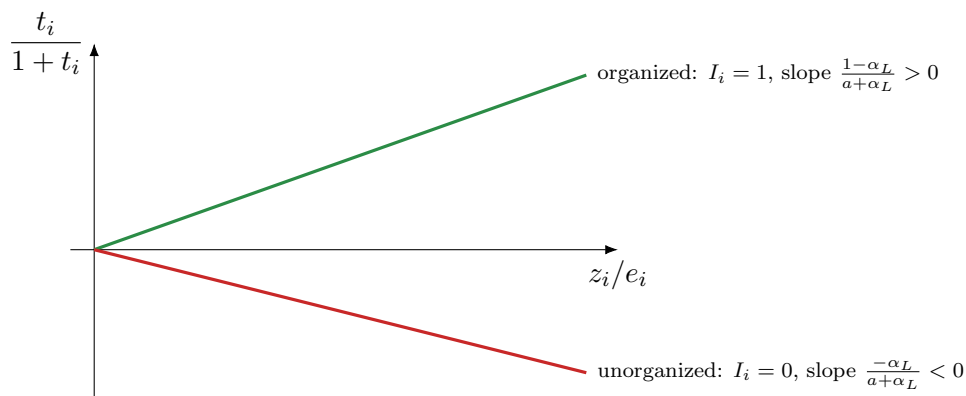
In any interior truthful Nash equilibrium,

$$\frac{t_i}{1 + t_i} = \frac{I_i - \alpha_L}{a + \alpha_L} \cdot \frac{z_i}{e_i}$$

where $I_i = 1$ if sector i is organized (else 0), α_L is the population share represented by any lobby, $z_i = y_i/m_i$ is the output-to-imports ratio, and e_i the import-demand (or export-supply) elasticity.

Reading the formula.

- A **modified inverse-elasticity (Ramsey) rule**: protection falls with e_i — the deadweight loss of a wedge rises with the trade elasticity, and the players internalize it whenever $a > 0$. The same elasticity logic as every optimal-tariff formula in Parts I–V.
- **Organized sectors are protected** ($t_i > 0$); **unorganized sectors are taxed** ($t_i < 0$): the numerator $I_i - \alpha_L$ is positive only for lobby members.
- **High z_i (big output, small trade volume) means high protection**: much to gain per specific-factor owner, little trade to distort — Mayer’s concentrated-benefit logic reappearing inside a lobbying model.
- **Limiting cases**: if everyone is organized ($\alpha_L = 1$, all $I_i = 1$), lobbies neutralize one another and **free trade** re-emerges; if $\alpha_L = 0$, unorganized sectors get exactly $t_i = 0$.



13.3 Why lobbies may prefer inefficient instruments

Grossman–Helpman, Section V

Even though a production subsidy is more efficient than a tariff per dollar transferred, organized lobbies may prefer that the government *not* have access to it: when rival lobbies compete intensely, an efficient instrument strengthens the government’s credible threat of a cheap deal with the rivals, raising the contributions each lobby must pay. Constraining

the government to *inefficient* instruments (tariffs, quotas, VERs — rents even accruing to foreigners) protects the lobby's bargaining position: a rigorous common-agency version of Bhagwati's DUP intuition for why real-world protection takes crude, high-deadweight-loss forms.

13.4 Synthesis: two routes to one conclusion

	Mayer (1984)	Grossman–Helpman (1994)
Political mechanism	Majority voting by factor owners	Campaign contributions / common agency
Unit of analysis	Individual factor owner's optimal tariff	Organized sector (lobby)
Source of bias	Skewed ownership; participation costs	Endogenous weights ($1 + a$) vs. a
Key result	Small organized sectors win protection despite minority status	$\frac{t_i}{1+t_i} = \frac{I_i - \alpha_L z_i}{a + \alpha_L e_i}$: organized protected, unorganized taxed
Relation to Parts I–V	Why observed tariffs $\neq$ welfare-maximizing ones	Testable cross-industry formula nesting the same elasticity logic

Both papers formalize the same insight — **concentrated benefits and diffuse costs generate a political bias toward protection** unrelated to terms-of-trade or profit-shifting motives — through two different institutional lenses: the ballot box and the campaign contribution.

Understanding Quiz 13 — Lesson 13

- Q13.1** In Protection for Sale, the government's objective $\sum_{i \in L} C_i + aW$ means it values:
- (A) only welfare
 - (B) contributions and welfare — equivalently, lobby members at weight $1 + a$ and others at weight a
 - (C) only contributions
 - (D) exports only
- Q13.2** According to the equilibrium formula, protection is higher for sectors that are:
- (A) organized, with high output-to-import ratios and low trade elasticities
 - (B) unorganized, trade-intensive, and elastic
 - (C) capital-intensive always
 - (D) export-oriented always
- Q13.3** If *every* citizen belongs to some lobby ($\alpha_L = 1$, all $I_i = 1$):
- (A) protection is maximal
 - (B) free trade emerges — rival lobbies fully neutralize one another
 - (C) the government collapses
 - (D) tariffs equal $1/e_i$
- Q13.4** Lobbies may prefer the government to use *inefficient* instruments because:
- (A) they misunderstand economics
 - (B) inefficiency weakens the government's threat of a cheap deal with rival groups, lowering the contributions needed to sustain the lobby's preferred policy
 - (C) tariffs are easier to spell

(D) efficient instruments are unconstitutional

Final Exam: Comprehensive Multiple-Choice Quiz

Final Exam — all parts

- F1.** A tariff in a small country reduces welfare by exactly:
- (A) $a + c$
 - (B) the two deadweight triangles $b + d$
 - (C) tariff revenue
 - (D) zero
- F2.** The large-country welfare decomposition is $\Delta W = -b - d + e$, where e is:
- (A) quota rent
 - (B) the terms-of-trade gain $(P^w - P^{t*})M$, extracted from foreign exporters
 - (C) consumer surplus
 - (D) the production distortion
- F3.** Johnson's optimum tariff is $t^{opt} = 1/\varepsilon^*$: a small country's optimal tariff is zero because:
- (A) it has no imports
 - (B) it faces an infinitely elastic foreign offer curve
 - (C) its government is benevolent
 - (D) of WTO rules
- F4.** Tariff-quota equivalence fails under domestic monopoly because:
- (A) quotas raise more revenue
 - (B) a quota leaves the monopolist a downward-sloping residual demand to exploit; a tariff only raises its price ceiling
 - (C) tariffs are ad valorem
 - (D) quotas are permanent
- F5.** Strategic substitutes vs. complements are defined by the sign of:
- (A) $\pi_{a_1}^1$
 - (B) the cross-partial $\pi_{a_1 a_2}^1$
 - (C) the determinant Δ
 - (D) the intercept
- F6.** In Cournot, a home tariff (a rise in the foreign rival's cost c_2) causes:
- (A) $X_1 \uparrow$, $X_2 \downarrow$, price $\uparrow$: profit shifts to the home firm
 - (B) both outputs rise
 - (C) price falls
 - (D) $X_1 \downarrow$
- F7.** Starting from free trade in the single-market Cournot model, a *marginal* tariff:
- (A) raises welfare by profit shifting
 - (B) lowers welfare: $dU|_{t=0} = -X_2 dt < 0$ — profit shifting is second-order, the consumer loss first-order
 - (C) has zero effect

(D) is prohibitive

F8. Johnson's answer to "must both countries lose a tariff war?" is:

- (A) yes, always
- (B) no: at most one must lose; relative offer-curve elasticities decide who can win
- (C) both always win
- (D) trade goes to zero

F9. Mayer's (1981) locus $t + t^* + tt^* = 0$ characterizes:

- (A) Nash tariffs
- (B) Pareto-efficient policy pairs: free trade, or one tariff matched by one subsidy
- (C) revenue-maximizing tariffs
- (D) the prohibitive tariff

F10. Syropoulos (2002): a country wins the bilateral tariff war when:

- (A) it is more patient
- (B) its relative size exceeds a unique threshold $\bar{\lambda}^i$
- (C) it moves second
- (D) its tariff is bound

F11. Brander–Spencer (1984): against a foreign monopolist, the optimal import policy is a subsidy when:

- (A) demand is linear
- (B) demand is sufficiently convex ($R < -1$, e.g. constant elasticity), so tariff pass-through exceeds one
- (C) the monopolist is small
- (D) never

F12. Brander–Spencer (1985): the optimal export subsidy makes the home firm behave like:

- (A) a price-taker
- (B) a Stackelberg quantity leader
- (C) a monopsonist
- (D) a colluder

F13. Under Bertrand competition the Brander–Spencer prescription flips to an export tax because:

- (A) prices are sticky
- (B) best responses slope up: committing to a higher price induces the rival to raise its price too, softening competition
- (C) taxes raise revenue
- (D) quantity is fixed

F14. The exporters' subsidy war (Propositions 4–5) is a Prisoner's Dilemma because:

- (A) subsidies are irreversible
- (B) each exporter subsidizes in equilibrium, yet both would gain if subsidies were jointly cut (the joint optimum is even an export tax)
- (C) the importer always loses
- (D) there is no equilibrium

F15. Mayer (1984): with skewed capital ownership and majority voting, the equilibrium tariff is set by:

- (A) the mean capital owner
- (B) the median voter, whose endowment ratio lies below the mean
- (C) the largest capitalist

(D) economists

F16. In Protection for Sale, unorganized sectors receive:

(A) high tariffs

(B) negative protection ($t_i < 0$): they are effectively taxed to fund organized sectors' protection

(C) free trade exactly

(D) quotas

Answer Key

Lesson quizzes

Lesson 1 — Q1.1: B (production and consumption distortions). Q1.2: B. Q1.3: B (revenue = rate $\times$ volume; one factor is zero at each end). Q1.4: B (no terms-of-trade channel).

Lesson 2 — Q2.1: B. Q2.2: B (first-order gain vs. second-order loss). Q2.3: B (t^{opt} inverse in ε^*). Q2.4: B (tangency of Home's trade IC with O_F^*).

Lesson 3 — Q3.1: B. Q3.2: B. Q3.3: B (rectangle lost to foreign license holders). Q3.4: B (price ceiling vs. residual demand curve).

Lesson 4 — Q4.1: B. Q4.2: B. Q4.3: B. Q4.4: B (tariff dominates quota under domestic monopoly).

Lesson 5 — Q5.1: B. Q5.2: A. Q5.3: B (envelope theorem kills the own-action term). Q5.4: B.

Lesson 6 — Q6.1: B. Q6.2: B. Q6.3: C (U-shape, then flat at exit). Q6.4: B.

Lesson 7 — Q7.1: B. Q7.2: B. Q7.3: B (two-way trade in the same good across segmented Cournot markets). Q7.4: B (second-best environment).

Lesson 8 — Q8.1: B. Q8.2: B. Q8.3: B. Q8.4: B (crossing vs. tangency of indifference curves).

Lesson 9 — Q9.1: B (exogeneity of the size measure). Q9.2: B. Q9.3: B. Q9.4: B.

Lesson 10 — Q10.1: B. Q10.2: B (pass-through > 1 when $R < -1$). Q10.3: B. Q10.4: B.

Lesson 11 — Q11.1: B. Q11.2: B. Q11.3: B. Q11.4: B. Q11.5: B ($t = s + s^*$).

Lesson 12 — Q12.1: B. Q12.2: B. Q12.3: B. Q12.4: B (Olson's logic, formalized).

Lesson 13 — Q13.1: B. Q13.2: A. Q13.3: B (lobbies neutralize). Q13.4: B.

Final exam

F1: B. Only $b + d$ is a net loss; a and c are transfers. **F2:** B. The tariff pushes the foreign supply price down; the revenue slice below P^w is paid by foreigners. **F3:** B. $\varepsilon^* \rightarrow \infty \Rightarrow t^{opt} \rightarrow 0$. **F4:** B. Quantity restrictions restore the monopolist's pricing power. **F5:** B. $\pi_{a_1 a_2}^1 < 0$: substitutes; > 0 : complements. **F6:** A. Cost pass-through in Cournot; the tariff is a rise in c_2 . **F7:** B. Envelope logic at $t = 0$. **F8:** B. Johnson contra Scitovsky; elasticities j, k decide. **F9:** B. Tangency of tariff-indifference curves; both-tariff pairs are never efficient. **F10:** B. Propositions 2 and 4. **F11:** B. $\hat{t} \geq 0$ as $R \geq -1$; constant

elasticity gives $R = -1 - 1/\varepsilon < -1$. **F12: B.** The subsidy replicates the leader's FOC via the rival's reaction slope. **F13: B.** Strategic complements reverse the profitable direction of commitment. **F14: B.** Both subsidize; joint welfare would rise by cutting; cooperative optimum is a tax — not self-enforcing. **F15: B.** Median $<$ mean under right skew; the median voter is labor-abundant. **F16: B.** $I_i = 0 \Rightarrow$ numerator $-\alpha_L < 0$.

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